

Project Design Phase-II Technology Stack (Architecture & Stack)

Date	05 July 2026
Team ID	
Project Name	Visualization Tool for Electric Vehicle Charge and Range Analysis
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Order processing during pandemics for offline mode

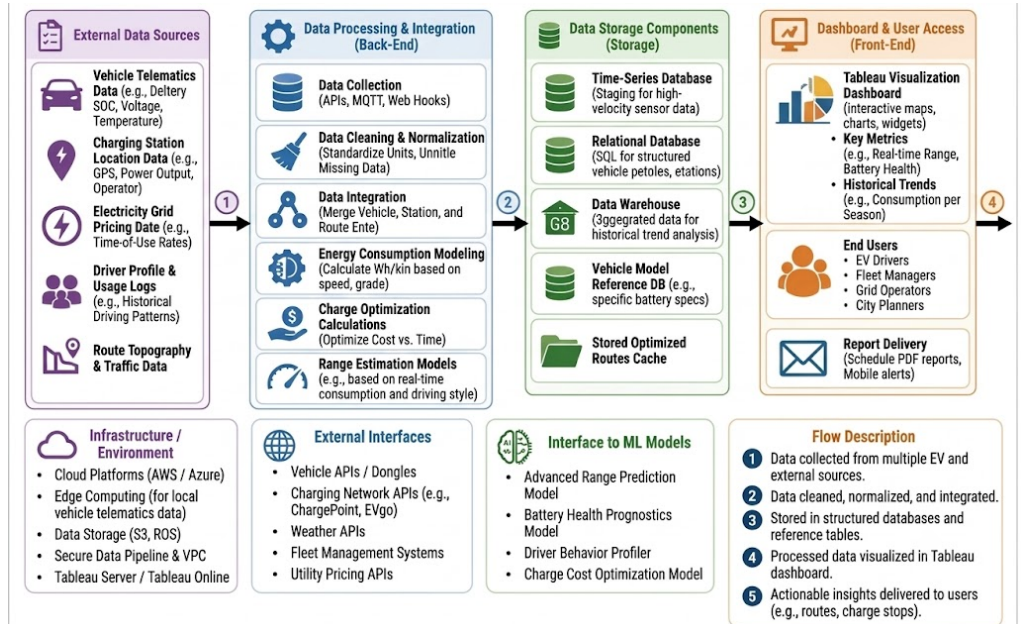
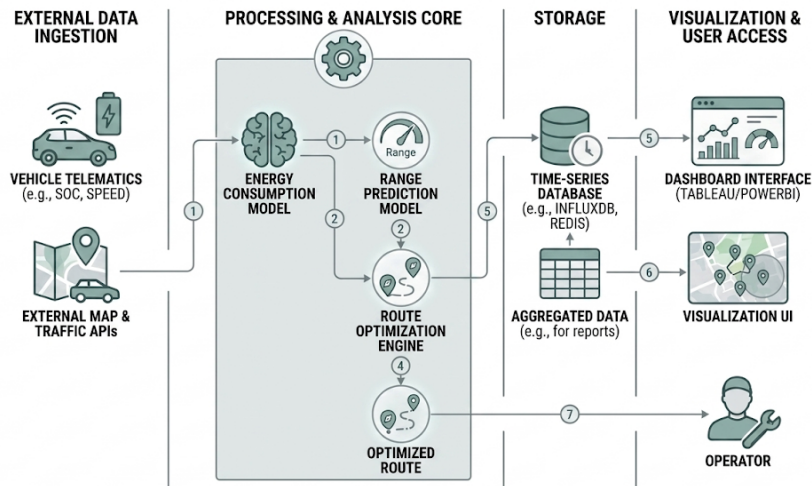


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	Web Dashboard & Mobile Application for real-time visualization of charge level, range, and topography, e.g., EV Driver UI, Fleet Manager UI.	React.js, Angular, React Native, Tableau/PowerBI embedded components.
2	Application Logic-1 (Backend)	Core services for vehicle data ingestion, range modeling, and route optimization.	Python (FastAPI / Flask), Java (Spring Boot), e.g., for range calculation.
3	Application Logic-2 (Streaming)	Real-time processing of live telematics streams from connected EVs for SOC (State of Charge).	IBM Event Streams, Apache Kafka, e.g., for processing live telemetry.
4	Application Logic-3 (Intelligence)	Predictive model deployment for range forecasting and degradation analysis.	IBM Watson Machine Learning, Python scikit-learn, e.g., for deployment of models.
5	Database (Primary)	Structured storage for user accounts, vehicle profiles, charging network	PostgreSQL, MySQL, with e.g., PostGIS for geospatial data.
6	Cloud Database (Analytical)	Scalable, high-throughput storage for massive amounts of historical trip telematics data for trend analysis.	IBM Cloudant (CouchDB), Cassandra, IBM DB2, for high-velocity IoT data.
7	File Storage	Centralized storage for large-scale route topography maps and historical weather grid data.	IBM Cloud Object Storage, AWS S3, or e.g., local files for testing.
8	External API-1 (Environment)	Integration with real-time weather services to incorporate temperature and wind into range calculations.	OpenWeatherMap API, IBM Weather API, e.g., for atmospheric conditions.
9	External API-2 (Infrastructure)	Integration with charging network APIs to provide live charger status, connector types, and pricing.	Open Charge Point Interface (OCPI), specific CPO APIs, e.g., for charger availability.
10	Machine Learning Model	Core neural network for precise SOC prediction and terrain-aware range forecasting.	Recurrent Neural Networks (RNNs) for range, Random Forests for health analysis, e.g., Battery SOC Prediction Model.

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	List the open-source frameworks used (e.g., for front-end charts, back-end API)	e.g. React.js, Python FastAPI, specific charting libraries.
2	Security Implementations	List all the security / access controls implemented, use of firewalls etc.	e.g. SHA-256 for data integrity, HTTPS/TLS, OAuth 2.0 for access, OWASP top 10 compliant libraries.
3	Scalable Architecture	Justify the scalability of architecture (3-tier, Micro-services) (e.g., handling increasing numbers of vehicle data points)	e.g. Micro-services using Kubernetes and Docker.
4	Availability	Justify the availability of application (e.g. use of load balancers, distributed servers etc.)	e.g. AWS ELB, content delivery networks.
5	Performance	Design consideration for the performance of the application (number of requests per sec, use of Cache, use of CDN's) etc.	e.g. Redis for caching frequently accessed map tiles or battery status.